

EARLY DIABETES DETECTION USING MACHINE LEARNING



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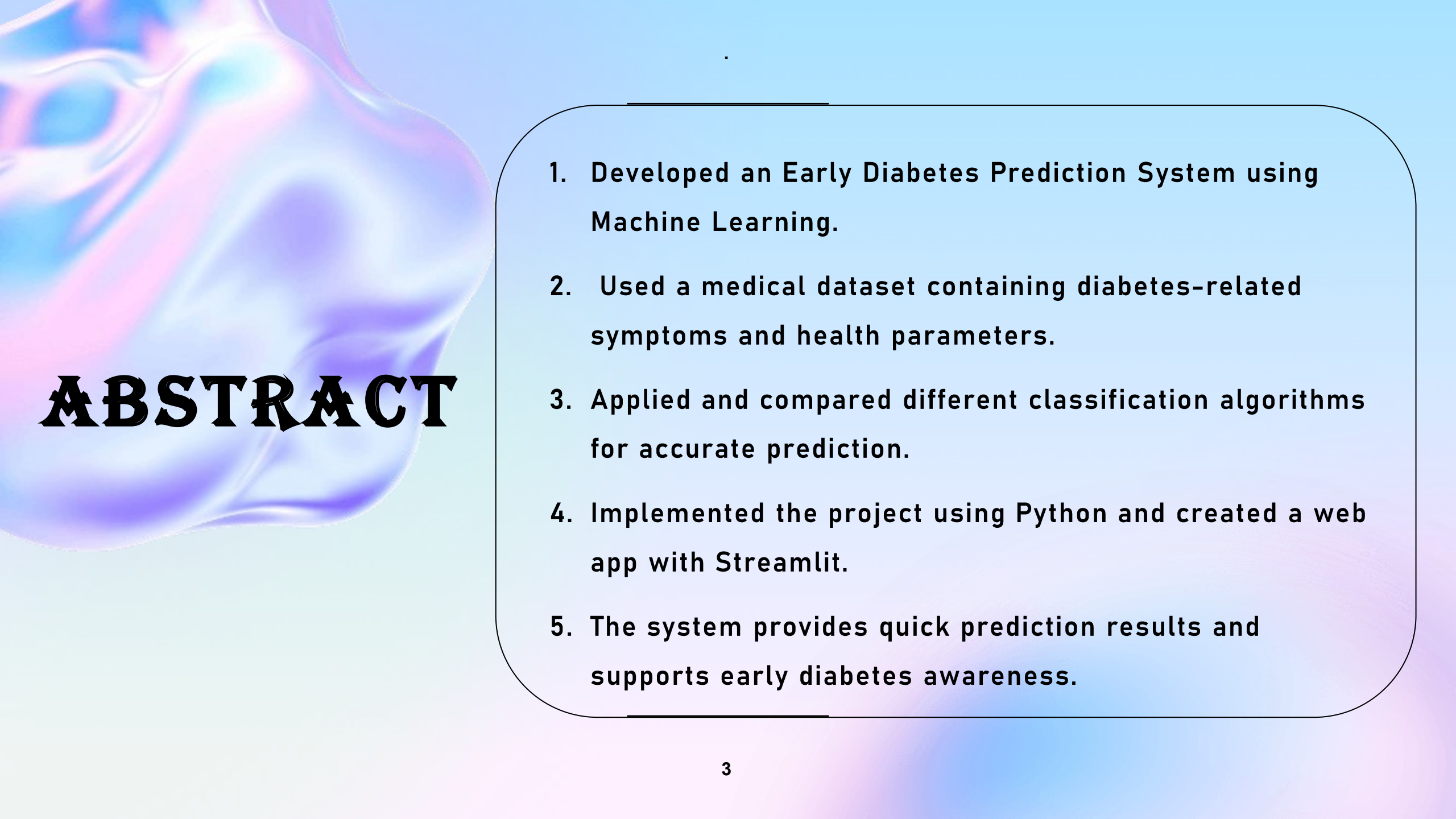
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ABSTRACT

1. Developed an Early Diabetes Prediction System using Machine Learning.
2. Used a medical dataset containing diabetes-related symptoms and health parameters.
3. Applied and compared different classification algorithms for accurate prediction.
4. Implemented the project using Python and created a web app with Streamlit.
5. The system provides quick prediction results and supports early diabetes awareness.

INTRODUCTION

1. Diabetes is a rapidly growing global health problem affecting millions of people.
2. Early symptoms are often ignored, leading to serious complications like heart, kidney, and eye diseases.
3. Major types of diabetes include Type 1, Type 2, and Gestational Diabetes. Traditional diagnostic methods are accurate but time-consuming and invasive.
4. Machine Learning helps in early diabetes prediction using medical data and symptoms.
5. ML-based systems provide faster, cost-effective, and accurate healthcare solutions.
6. This project focuses on developing an intelligent diabetes prediction system using Machine Learning and Python.

PROBLEM STATEMENT

EARLY DETECTION CHALLENGE

- Diabetes is increasing rapidly worldwide and early diagnosis is very important.
- Delayed detection can lead to heart, kidney, nerve, and eye complications.

Limitations of Traditional Methods

- Laboratory tests are time-consuming and costly.
- Traditional methods may fail to identify early-stage risk patterns.

NEED FOR INTELLIGENT PREDICTION SYSTEM

- Manual analysis of symptoms is difficult and complex.
- An automated Machine Learning system is needed for accurate prediction.

Challenges in Prediction

- Medical datasets may be imbalanced and complex.
- Different symptoms have different levels of importance.

Aims And Objectives

Aim

To develop a Machine Learning-based system for early prediction of diabetes using clinical and lifestyle data.

Objectives

1. ■ To collect and preprocess diabetes-related medical data.
2. To implement and compare different Machine Learning algorithms.
3. To improve prediction accuracy using preprocessing and balancing techniques.
4. To evaluate model performance using metrics like accuracy, precision, recall, and F1-score.
5. To identify the most effective model for early diabetes prediction.
6. To develop a reliable and user-friendly diabetes prediction system.

LITERATURE REVIEW

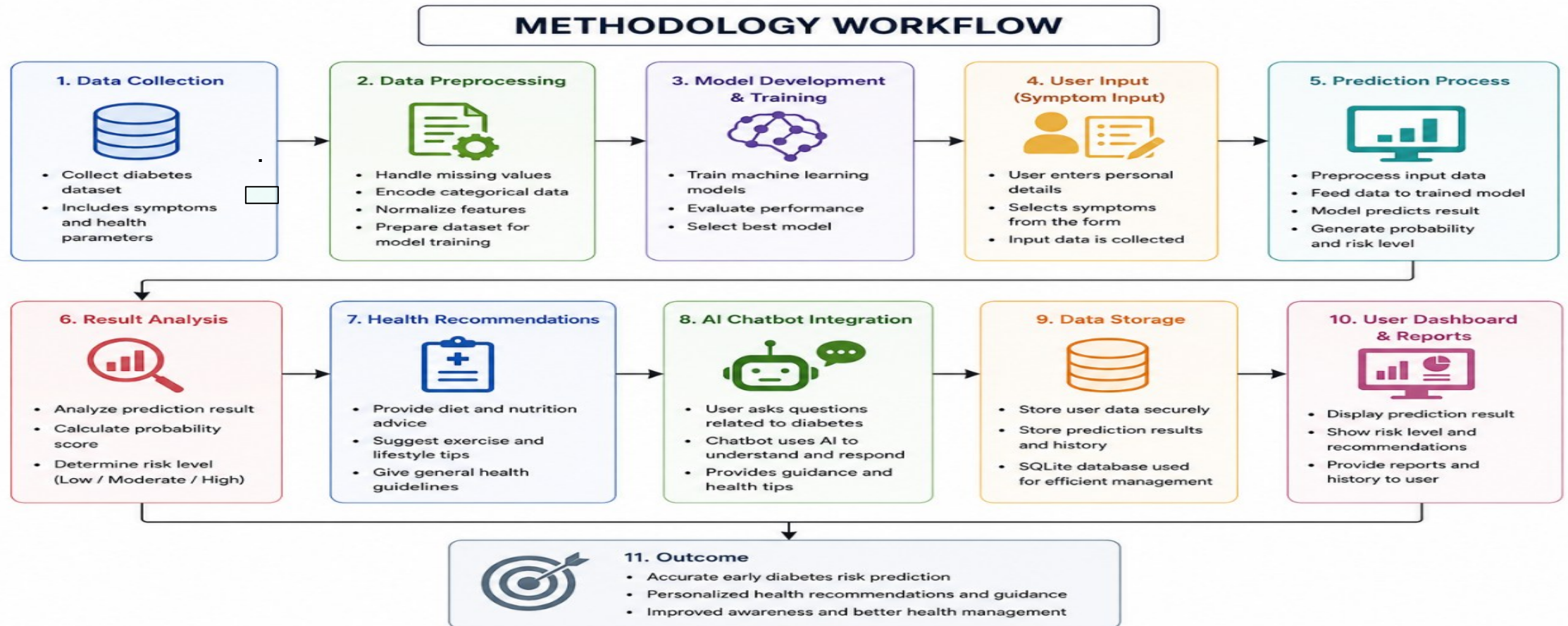
Authors	Year	Method Used	Findings	Limitations
D. Sisodia & D. S. Sisodia	2018	Classification Algorithms	Compared ML algorithms for diabetes prediction.	Depends on dataset quality.
I. Kavakiotis et al.	2017	Machine Learning & Data Mining Methods	Reviewed ML methods in diabetes research.	Limited practical implementation.
Q. Zou et al.	2018	Machine Learning Techniques	Improved diabetes prediction accuracy.	Needs large balanced datasets.
S. Perveen et al.	2016	Data Mining Classification Techniques	Evaluated different classifiers for diagnosis.	High computational complexity.
A. Mujumdar & V. Vaidehi	2019	Machine Learning Algorithms	Proposed efficient diabetes prediction models.	Accuracy varies across datasets.
J. J. Khanam & S. Y. Foo	2021	Comparative ML Algorithms	Compared performance of multiple algorithms.	Overfitting on limited data.

RESEARCH GAP

- Existing diabetes prediction systems mainly focus on basic prediction methods with limited accuracy.
- Many studies do not handle imbalanced datasets and missing values effectively.
- Some models fail to identify diabetes at an early stage using symptom-based data.

- Limited comparison of multiple Machine Learning algorithms in a single system.
- Lack of user-friendly and real-time prediction systems for non-technical users.
- Need for more accurate, reliable, and interpretable prediction models in healthcare.

METHODOLOGY WORKFLOW



DATASET

CLINICAL FEATURES

1. AGE
2. Gender
3. Obesity
4. Weakness
5. Irritability
6. Delayed Healing
7. Partial Paresis
8. Alopecia
9. Class Label

SYMPTOMS

1. Polyuria
2. Polydipsia
3. Sudden Weight Loss
4. Polyphagia
5. Visual Blurring
6. Itching
7. Muscle Stiffness
8. Genital Thrush

DATA PROCESSING

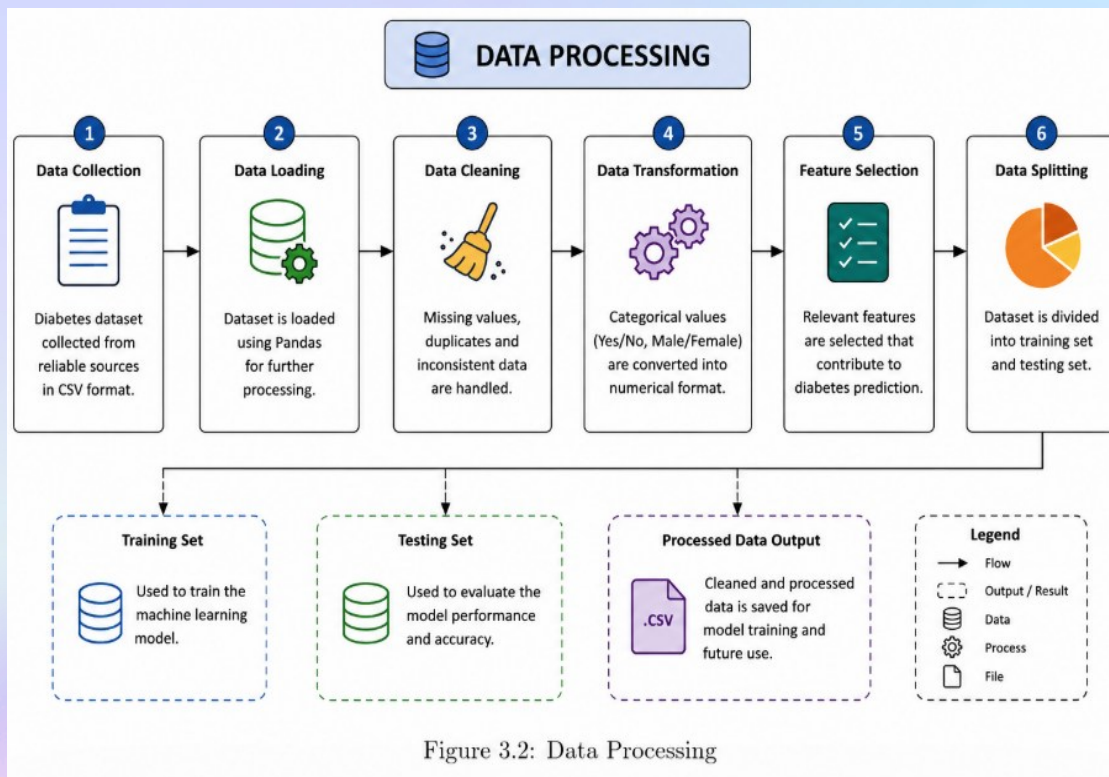


Figure 3.2: Data Processing

- Checked dataset for missing values, duplicate records, and inconsistencies.
- Removed duplicate entries to improve data accuracy. Converted categorical values like “Yes/No” and “Positive/Negative” into numerical form using Label Encoding.
- Performed feature analysis to identify important attributes for diabetes prediction. Split the dataset into training and testing data for model building and evaluation.
- Applied preprocessing and feature transformation techniques to prepare data for machine learning models.

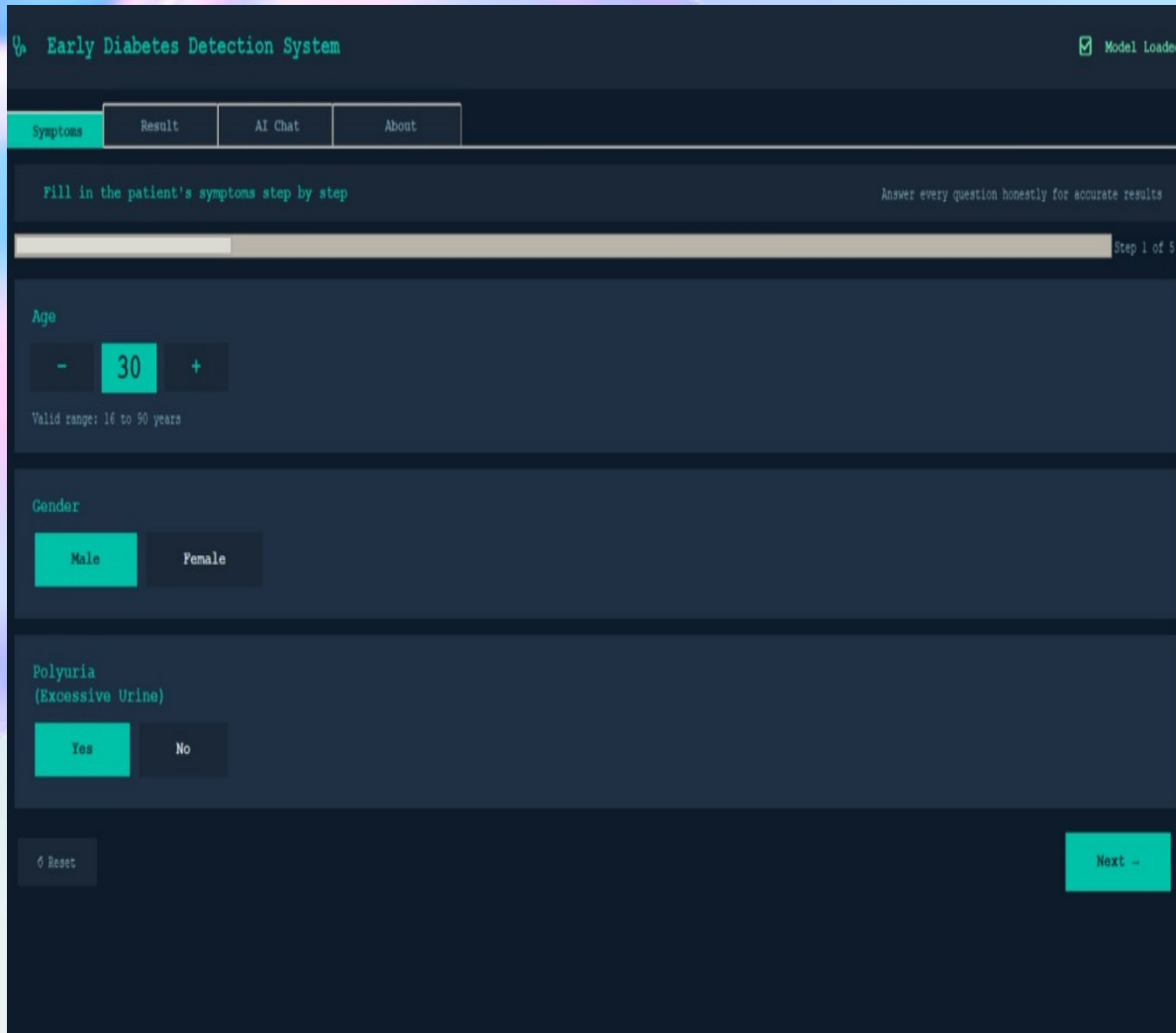
FEATURE SELECTION

- Used statistical analysis and **exploratory data analysis (EDA)** to study feature importance.
- Applied the Chi-Square test to measure the relationship between features and the target variable. Important features included **polyuria, polydipsia, sudden weight loss, weakness, visual blurring, and age**.
- Features with p-value less than 0.05 were considered significant and retained. Reduced unnecessary features to improve model efficiency, accuracy, and reliability.

MACHINE LEARNING MODEL DEVELOPMENT

- Divided the dataset into training and testing data. Trained different machine learning models for diabetes prediction.
- Used selected features from the feature selection process for model training.
- Applied multiple algorithms such as **Logistic Regression, KNN, Decision Tree, Random Forest, SVM, Naive Bayes, AdaBoost, and XGBoost**.
- Evaluated models using **accuracy, precision, recall, F1-score, and confusion matrix**.
- Random Forest and XGBoost showed strong performance,
- SVM achieved: **Accuracy:** 96%, **Precision:** 98–99% , **Recall Value:** Non-diabetic class (0): 94% Diabetic class (1): 97% , **F1-Score:** 95–97%

System Design & User Interface Development



The screenshot displays the 'Early Diabetes Detection System' interface. At the top, there's a title bar with a home icon and the text 'Early Diabetes Detection System', and a status indicator 'Model Loaded' with a checkmark. Below the title bar is a navigation menu with four tabs: 'Symptoms' (active), 'Result', 'AI Chat', and 'About'. The main content area is titled 'Fill in the patient's symptoms step by step' with a sub-note 'Answer every question honestly for accurate results'. A progress bar indicates 'Step 1 of 5'. The first question is 'Age', with a numeric input field showing '30' and a range 'Valid range: 16 to 90 years'. The second question is 'Gender', with radio buttons for 'Male' (selected) and 'Female'. The third question is 'Polyuria (Excessive Urine)', with radio buttons for 'Yes' (selected) and 'No'. At the bottom, there are 'Reset' and 'Next >' buttons.

- Developed an interactive and user-friendly diabetes prediction system using Python.
- Used machine learning models for diabetes risk prediction based on patient symptoms.
- Designed the frontend using the Tkinter library for an easy and attractive interface.
- Implemented a modular system design for better maintenance and future improvements.
- Added features such as symptom input, data preprocessing, prediction generation, and result visualization.

RESULT ANALYSIS & HEALTH RECOMMENDATIONS

- The system predicts whether a user is diabetic or non-diabetic based on entered symptoms.
- Generated a probability score to show prediction confidence and diabetes risk level.
- Classified results into low, moderate, and high-risk categories using visual indicators and color highlights.
- Symptoms like excessive thirst, frequent urination, weakness, and weight loss strongly affected prediction results.
- Provided personalized health recommendations for improving lifestyle and reducing diabetes risk. Recommendations included healthy diet, exercise, water intake, stress management, blood sugar monitoring, and regular medical check-ups.

RESULT ANALYSIS & HEALTH RECOMMENDATIONS

AI CHATBOT

🔍 Early Diabetes Detection System

Model Loaded

Symptoms

Result

AI Chat

About

⚠️ Diabetic

Diabetes Probability: 85%

Risk Level: HIGH RISK

📌 Recommended Actions for Diabetic Patients

Foods to EAT

1: Non-starchy vegetables (broccoli, spinach)

2: Berries – low glycemic index

3: Fatty fish (salmon, mackerel)

4: Eggs – great protein, low carbs

5: Legumes (lentils, chickpeas, rajma)

6: Whole grains (brown rice, oats)

7: Avocado – healthy fats & fiber

8: Citrus fruits (in moderation)

9: Almonds, walnuts, chia seeds

10: Green tea / fenugreek

Foods to AVOID

1: Sugary drinks, sodas, juices

2: White bread, maida, refined flour

3: White rice in large quantities

4: Deep-fried fast food

5: Sweets, mithai, cakes, biscuits

6: Full-fat dairy in excess

7: Alcohol

8: Packaged fruit juices

9: Ultra-processed snacks

Lifestyle Tips

1: Walk 30 min daily (after meals)

2: Take medication on time

3: Monitor blood sugar regularly

4: Yoga or meditation daily

5: Drink 8-10 glasses of water

6: Get 7-8 hours of sleep

7: Doctor check-up every 3 months

8: Quit smoking

9: Maintain healthy body weight

10: Eat small meals every 3-4 hours

🔍 Early Diabetes Detection System

Model Loaded

Symptoms

Result

AI Chat

About

AI API Key: Free key – openrouter.ai/keys

🗨️ DiabetesBot is ready!

Ask anything about diabetes, diet, symptoms, or your results.

You: What should I eat?

Bot: Focus on lean protein, non-starchy vegetables, and whole grains. Keep sugary drinks and refined carbs to a minimum.

You: How to control sugar?

Bot: Eat balanced meals with fiber and protein. Limit processed foods and added sugars. Stay hydrated with water. Small changes make a big difference!

You: Is diabetes curable?

Bot: None

You: What is Polyuria?

Bot: Polyuria means you urinate more than usual. It can be a symptom of diabetes or other conditions. Consult a doctor for proper diagnosis and management.

Quick: What is diabetes? What should I eat? How to control sugar? What is Polyuria? Is diabetes curable?

Send ➡

AI CHATBOT INTEGRATION

- Integrated an AI-powered chatbot to provide diabetes-related guidance and support.
- Built using Python with OpenRouter AI integration for interactive communication.
- Allowed users to ask questions about diabetes symptoms, diet, lifestyle, and health management.
- Generated short, simple, and user-friendly responses in real time.
- Improved user engagement by providing instant healthcare assistance and recommendations.

PLAN OF WORK

DATA COLLECTION & PREPROCESSING (DECEMBER – JANUARY)

Collected Early Diabetes Prediction dataset. Performed data cleaning, label encoding, and feature selection. Analyzed important symptoms related to diabetes prediction.

MODEL DEVELOPMENT & TRAINING (FEBRUARY)

Developed machine learning models such as SVM, Logistic Regression, Random Forest, KNN, and XGBoost. Evaluated models using accuracy, precision, recall, and F1-score

SYSTEM INTEGRATION (MARCH)

Integrated preprocessing and prediction modules into a single system. Implemented symptom input, prediction generation, and result analysis. Added health recommendation.

USER INTERFACE & FINAL IMPLEMENTATION (APRIL)

Developed an interactive graphical user interface using Tkinter. Added result visualization with probability and risk level indicators. Integrated chatbot for diabetes guidance.

FUTURE ENHANCEMENTS

- Use larger medical datasets to improve prediction accuracy.
- Implement deep learning algorithms for advanced analysis.
- Develop web and mobile application versions for easy accessibility.
- Add real-time patient monitoring and cloud database support.
- Integrate wearable devices like smartwatches and fitness trackers.
- Include prediction modules for additional diseases.
- Improve chatbot intelligence using advanced NLP techniques.
- Add multi-language support for better user accessibility.

CONCLUSION

- Developed an intelligent and user-friendly **Early Diabetes Detection System using Machine Learning and Python**.
- The system predicts diabetes risk based on **symptoms and medical information**. Applied and compared multiple machine learning algorithms to achieve accurate prediction results.
- Evaluated the model using **accuracy, precision, recall, and F1-score metrics**. Designed an interactive graphical user interface using **Tkinter** for easy symptom input and result visualization.
- Provided **health recommendations, lifestyle guidance, and AI chatbot** support for better user interaction.
- The project demonstrates the effective use of machine learning in healthcare for early disease prediction and awareness.
- The system can assist users in preliminary diabetes risk assessment, although it is not a replacement for professional medical diagnosis.

The background features a soft, pastel-colored gradient of blue, purple, and pink. A large, faint, smiling face is visible in the background. On the right side, there are decorative elements: a white starburst icon near the top and a white wavy line near the bottom.

THANK YOU

FOR YOUR TIME AND ATTENTION

Presented By:

Tanisha And Nidhi Verma